ONE DISTRICT ONE PRODUCT
UTTAR PRADESH

A GUIDE TO PRODUCT QUALITY STANDARDS

METAL CRAFT

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INDEX

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ABBREVATIONS

ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
AS	Australian Standards
ASTM	American Society for Testing and Materials
BICON	Bio Security Import Condition System
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPSA	Consumer Product Safety Act
DOT	Department of Transportation
EC	Council Regulation (Regulation of the European Parliament and Council)
ECHA	European Chemicals Agency
EN	European Standards
EU	European Union
FDA	Food and Drug Administration
FHSA	Federal Hazardous Substances Act
FSSAI	Food Safety and Standards Authority of India
GCC	Gulf Co-operation Council
GSO	GCC Standardization Organization
GPSD	General Product Safety Directive
IS	Indian Standard
ISI	Indian Standards Institution
ILAC	International Laboratory Accreditation Cooperation
ISO	International Organization for Standardization
MLA	Multilateral Recognition Arrangement
MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RoHS	Restriction of Hazardous Substances Directive
SASO	Saudi Standards, Metrology and Quality Organization
SRL	Specific Release Limits
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This compendium brings relevant regulations and standards pertaining to the metal craft industry and goods. Uttar Pradesh is famed for its metal crafts which are created with different metals that include household items like pots, trays, bowls and decorative pieces with intricate etching. Moradabad and Varanasi are well known for the production of metal artefacts in the state with Mirzapur emerging in recent times as important centres of production.

BELLS & GHOONGROO, MIRZAPUR, ETAH, MAHOBA

1. INTRODUCTION

Historically, bells have been associated with religious rites for centuries. It was only later, that bells were made to commemorate important events or people and have been associated with the concepts of peace and freedom. Further, in almost all Hindu temples there are one or more bells hung from the top near the entrance.

Bells are usually cast from bell metal (a type of bronze) for its resonant properties, but can also be made from other hard materials. This depends on the function of the bell. Some small bells such as ornamental bells or cowbells can be made from cast or pressed metal, glass or ceramic, but large bells such as a church, clock and tower bells are normally cast from bell metal.

A bell is a directly struck idiophone percussion instrument. Most bells have the shape of a hollow cup that when struck, vibrate in a single strong strike tone with its sides forming an efficient resonator. The strike may be made by an internal "clapper" or "uvula", an external hammer, or—in small bells—by a small loose sphere enclosed within the body of the bell. Bells intended to be heard over a wide area can range from a single bell hung in a turret or bell-gable, to a musical ensemble such as an English ring of bells, a carillon or a Russian zvon which are tuned to a common scale and installed in a bell tower. Many public or institutional buildings house bells, most commonly as clock bells to sound the hours and quarters.

The Bell makers of Etah are renowned the world over for their expertise in making bells. They regularly receive orders for bells from different parts of the world. A testament of their skill is the large 2,100 Kg. bell being cast for the Ram Mandir at Ayodhya.

Further, Mirzapur and Mahoba are well known for their brassware industry. The craftsmen are skilled in casting several exquisite forms using a wide variety of metals and alloys. These products range from sculptures to religious iconography.

2. RAW MATERIALS

Main constituent of bell alloy are copper and tin which are relatively soft metals that deform on striking (though tin to a lesser extent than copper). However, alloying them creates a metal (Bronze) which is harder and less ductile and also one with more elasticity than either of the two metals. This metal combination produces a tough, long-wearing material that is resistant to oxidation and subject only to an initial surface weathering. Specifically, it is the combination of low internal damping and low internal sound velocity that makes bell metal especially suitable for resonant percussion instruments. Most commonly, as per its colloquial name, bell metal has been and is still used for the casting of high-quality bells. The metal used is a high-tin alloy of copper and tin with approximately a 4:1 ratio of copper and tin (78% copper, 22% tin). This is a much higher tin component than that used in statuary bronze.

The percentage of tin content can vary from 20 to 26%, depending on the founder who usually arrive empirically at their own alloy ratio. It has been found that increasing the tin content increases the decay time of the bell strike, thus making the bell more sonorous.

3. MANUFACTURING PROCESS OF BELL

The process of casting bells is called bell founding. The manufacturing process starts with calculating the bell design based on the specifications required and the ratio of alloys after which a template is made. The bells can be cast using the following methods:

- Bell foundry sand casting method
- Lost wax method

3.1. Bell foundry sand casting method

The following form the backbone of any bell foundry sand casting manufacturing processes:

a. **Casting**

- **External mould:** An exact model of the external or outer part of the bell is made using stone, brick or coke. This is covered with clayey loam. A total of three or more layers of loam may be applied. After each application of loam, the cast is dried in a kiln.
- **Internal mould:** Just as the external mould, a secondary mould is made for the internal bell.

b. **Pouring:** The two separate moulds are fitted together, leaving a gap in between the two. Molten metal conforming to the specifications determined previously is poured in and allowed to solidify.

c. **Cooling:** The bell mould is carefully lowered into a casting pit in the ground and the metal is allowed to cool over for a few days.

Figure 1: Casting of a bell

3.2. Lost wax method

The lost wax bell casting method is one where a cast is made from an already existing bell. The following processes are generally employed:

a. **Model Making:** An exact replica of the existing bell is made using wax primarily, as it is an easy material to work with and is relatively inexpensive. Other materials such as an oil based clay may be used too.

b. **Mould Making:** A mould is made usually in two halves of the original wax model. The inner mould is the exact negative of the original model. Wax is then poured into this mould and swished around to coat the sides of the mould evenly.

c. **Spruing:** After the hollow wax halves are removed from the mould, they are finished by a process called chasing to remove the marks left by the mould. The two halves are heated and joined together. Foundries often use registration marks to indicate exactly where they would go. This wax copy is sprued with a tree like structure of wax that eventually provides paths for the molten casting metal to flow and for air to escape.

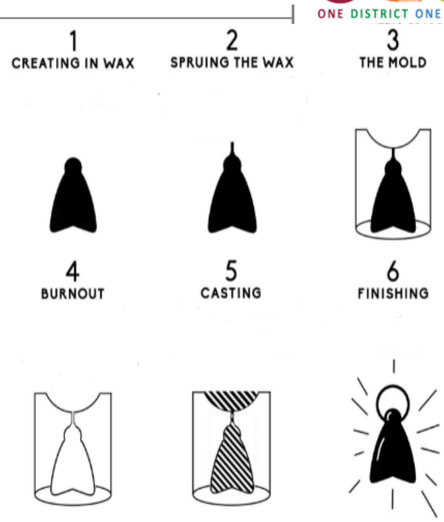

Figure 2: The steps in the lost wax method

d. **Slurry Coating:** The readied piece is dipped into a slurry of silica and then in a sand like stucco. This slurry and sand mixture is called a ceramic shell mould material (though it is not actually ceramic) in the industry parlance. This shell is allowed to dry and the process is repeated till the required thickness of the external coating has been achieved.

e. **Burnout:** The coated shell is heated in a kiln, which causes the wax inside to melt. This usually gets burnt up, lending its name to the process "Lost wax method". Only the negative space remains within the mould.

f. **Pouring:** The mould is heated again in the kiln and molten metal is poured in. It is important that there should be no temperature difference between the moulds and metal otherwise the mould would shatter.

g. **Finishing:** The metal is allowed to cool, and then the external mould is broken. The sprues which are attached to the cast shape are cut off and the entire object is finished by removal of all rough edges etc. and polished.

Great care is taken to calculate the precise weight and size of the bell before it is cast. If the finished bell does not meet specifications, it is completely melted down and recast. In case, a bell cracks at a future date, it might be welded and patched, but that is rare. The bell is more likely to be retired or it is melted down and recast.

Figure 3: Tuning a bell

Bells are casted uniquely to ensure certain amounts of metal can be removed to adjust the harmonics within. Removing these metals allows the bell-operator to accurately tune the bell and adjust the harmony. There are five core principals of bell tuning:

- Hum
- Fundamental
- Tierce
- Quint
- Nominal

After the desired shape and sound is achieved, polishing is done using chemicals (Bright dipping) and then anodizing/hard anodizing.

4. MANUFACTURING PROCESS OF GHOONGHROO

Ghoonghroos are a type of crotal bells, which can be made using a wide variety of materials but the most commonly used material is brass. These small bells are made as under:

4.1. Cutting the strip of brass

The brass strip is cut into a small square shape and cuts are made on the edges.

4.2. Inserting the bell insert

A small hole is drilled in the center to allow for inserting a thread, which can be used to fasten the bells on to a surface. A small ball of metal is placed at the centre

4.3. Shaping the bell

The metal sheet is bent to seal in the metal ball and using a needle, the bell is threaded.

When shaken, the free small ball of metal strikes the insides of the bell to produce a sound. More intricate crotal bells can be cast using the lost wax technique.

5. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check the fabricated bell for workmanship
- Check for the uniformity of shape and dimensions
- Check for finishing of anodizing coating and damages, if any
- Check for hairline cracks and sharp edges

6. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

6.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

6.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European union	a. Regulation (EU) 2015/628 of 22 April 2015 b. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulation (EC) 1907/2006	a. European Commission b. European Chemicals Agency (ECHA)	a. Specifies the percentages of Lead in the alloy and its migration b. Hazardous chemicals
2	United States of America	a. Consumer Product Safety Improvement Act	a. Consumer Products Safety	a. Specifies requirements with respect to Chemicals and heavy metals

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
		(CPSIA), Act of 2008 **b.** Consumer Product Safety Act (CPSA), Act of 1972 **c.** Federal Hazardous Substances Act (FHSA), Act of 1960. **d.** California Proposition 65, Act of 1986	Commission (CPSC) **b.** California State	**b.** Specifies requirement for suitable product labelling **c.** Specifies banned substances **d.** Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products
3	Australia	**a.** Commerce (Trade Description) Act, 1905	**a.** Australian Customs Service	**a.** Specifies the requirements for Mandatory Labelling of products
4	Middle East	**a.** Saudi Standards, Metrology and Quality Organization (SASO) Regulations.	**a.** Saudi Standards, Metrology an Quality Organization (SASO)	**a.** Specifies product safety requirements

7. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Scope**	**Region/ Country**	**Standard**	**Scope**
General (Refer Table 1)	**IS 928: 1984 (Reaffirmed Year: 2020)**	Specifications for fire bells	USA	ASTM B846 - 19a	Standard terminology for copper and copper alloys
	IS 1387: 1993 (Reaffirmed Year: 2018)	General requirements for the supply of metallurgical materials	Middle East	GSO ISO 197-1: 2015	Copper and copper alloys - Terms and definitions - Part 1: Materials
Raw Materials (Refer Table 2.a & 2.b)	**IS 3685: 1966 (Reaffirmed Year: 2018)**	Chemical analysis of brasses	European Union	BS EN 1981 - 2003	Copper and copper alloys. Master alloys
	-	-	European Union	BS EN 1655: 1997	Copper and copper alloys. Declarations of conformity
	-	-	USA	ASTM F956 – 91(2018)	Specification for bell, cast, sound signalling
	-	-	USA	ASTM B36	Standard specification for brass plate, sheet, strip, and rolled bar
	-	-	Australia	AS 1565-1996	Copper and copper alloys - Ingots and castings

Domestic Standards			International Standards		
Stage	**Code**	**Scope**	**Region/ Country**	**Standard**	**Scope**
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 292: 1983 (Reaffirmed Year: 2021)**	Specifications for leaded brass ingots and casting	European Union	BS EN 12420: 2014	Copper and copper alloys forging
	IS 306: 1983 (Reaffirmed Year: 2021)	Specifications for tin bronze ingots and casting	European Union	BS EN 1982: 2017	Copper and copper alloys. Ingots and castings
	IS 1408: 1968 (Reaffirmed Year: 2021)	Recommended procedure for inspection of copper-base alloy sand castings	USA	ASTM B584 – 14	Standard specification for copper alloy sand castings for general applications
	-	-	Australia	AS 1515.4-1978	Methods for the analysis of copper alloys, Part 4: Electrolytic determination of copper in wrought and cast copper alloys
Finishing (Refer Table 4.a & 4.b)	**IS 3658: 1999 (Reaffirmed Year: 2020)**	Code of practice for liquid penetrant flaw detection	Middle East	GSO ISO 4751: 2013	Copper and copper alloys - Determination of tin content - Spectrometric method
	IS 1501: Part 1: 2020	Metallic materials — Vickers hardness test , part 1 test method (Fifth revision)	Middle East	GSO ISO 3220: 2013	Copper and copper alloys - Determination of arsenic - Photometric method
	IS 4027: Part 1: 1987	Methods of chemical analysis of bronzes Part 1 Determination of copper and lead by electrolytic method	-	-	-

7.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 928: 1984 (Reaffirmed Year: 2020)** - Specifications for fire bells	▪ Requirements regarding materials, shape, finishing, dimensions and workmanship ▪ Adjunct standards and referenced documents
	IS 1387: 1993 (Reaffirmed Year: 2018) - General requirements for the supply of metallurgical materials	▪ General requirements for the supply of metallurgical materials ▪ Terminology ▪ Sampling methods
	ASTM B846 - 19a - Standard terminology for copper and copper alloys	▪ Copper alloy sand castings for general applications ▪ Nominal compositions ▪ Units of measurements
	GSO ISO 197-1: 2015 - Copper and copper alloys - Terms and definitions - Part 1: Materials	▪ Glossary of terms for copper and copper alloys

7.2. Table 2.a: (Domestic Regulations for Raw Materials)

Stage	Code	Scope
Raw Materials	**IS 3685: 1966 (Reaffirmed Year: 2018)** - Methods of chemical analysis of brasses	▪ Methods for determination of copper, lead, tin, manganese, phosphorus, nickel, iron, silicon, aluminium, zinc, and arsenic in the ranges as specified in the relevant Indian Standards on brasses. ▪ Quality of reagents ▪ Methodology ▪ Referenced documents

7.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**BS EN 1981 - 2003 -** Copper and copper alloys. Master alloys	▪ Specifies the compositions of copper-based master alloys intended for the manufacture, de-oxidation, or desulfurization of cast or wrought alloys, especially those based on copper, supplied in the form of ingots, notched bar, notched slab (waffle plate), granules or broken pieces ▪ Sampling of master alloys
	BS EN 1655: 1997 - Copper and copper alloys. Declarations of conformity	▪ Copper and copper alloys, Declarations of conformity
	ASTM F956 – 91 - Standard specification for bell, cast and sound signalling	▪ Cast sound-signalling bells, together with bulkhead mounting plates for the smaller bells, for use on ships, boats, and other marine craft, for compliance with the rules in the convention on the international regulations for preventing collisions at sea ▪ Units of measurements to be used ▪ Adjunct standards and referenced documents
	ASTM B36 - Standard specification for brass plate, sheet, strip, and rolled bars	▪ Requirements for brass plate, sheet, strip, and rolled bar of different alloys ▪ Units of measurements ▪ Referenced documents
	AS 1565-1996 - Copper and copper alloys - Ingots and castings	▪ Copper and copper alloys - Ingots and castings

7.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 292: 1983 (Reaffirmed Year: 2021) - Specification for leaded brass ingots and casting	▪ Requirements for Leaded brass ingots two grades LCB1 and LCB2 ▪ Chemical composition and tests ▪ Referenced documents and referee method of chemical composition test
	IS 306: 1983 (Reaffirmed Year: 2021) - Specification for reaffirmed tin bronze ingots and castings	▪ Requirements of tin and bronze casting ▪ Chemical compositions ▪ Referee methods
	IS 1408: 1968 (Reaffirmed Year: 2021) - Recommended procedure for inspection of copper base alloy sand casting	▪ Industrial inspection of copper base alloy sand casting ▪ Classifications of casting

7.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS EN 12420: 2014 - Copper and copper alloys –Forgings	▪ Copper and copper alloys-Forgings
	BS EN 1982: 2017 - Copper and copper alloys-Ingots and castings	▪ Chemical composition ▪ Mechanical properties ▪ Sampling and testing
	ASTM B584 – 14 - Standard specification for copper alloy sand castings for general applications	▪ Copper alloy sand castings for general applications ▪ Nominal compositions ▪ Units of measurements
	GSO ISO 197-1: 2015 - Copper and copper alloys - Terms and definitions - Part 1: Materials	▪ Glossary of terms

7.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 3658: 1999 (Reaffirmed Year: 2020) - Code of practice for liquid penetrant flaw detection	▪ Code of practice for liquid penetrant flaw detection ▪ Referenced documents ▪ Adjunct standards
	IS 1501-1: 2020 - Metallic materials — Vickers hardness test part 1 test method (Fifth revision)	▪ Vickers hardness test
	IS 4027: Part 1: 1987 - Method of chemical analysis of bronzes	▪ Determination of copper and lead by electrolytic methods ▪ Referenced documents

7.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	AS 1515.4-1978 - Methods for the analysis of copper alloys - Part 4: Electrolytic determination of copper in wrought and cast copper alloys	▪ Methods for the analysis of copper alloys - Part 4: Electrolytic determination of copper in wrought and cast copper alloys
	GSO ISO 4751: 2013 - Copper and copper alloys - Determination of tin content - Spectrometric method	▪ Tests for determination of tin by spectrometry
	GSO ISO 3220: 2013 - Copper and copper alloys - Determination of arsenic - Photometric method	▪ Tests for determination of arsenic by photometry

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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7. Mandatory IS mark : https://bis.gov.in/index.php/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/
8. Indian Standards under Compulsory Certification : https://www.bsbedge.com/mandatory-indian-standards
9. International Legislation/Regulation:
   a. EU : https://www.compliancegate.com/heavy-metal-regulations-european-union/ https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32015R0628&from=EN
   b. USA : https://www.compliancegate.com/heavy-metal-regulations-united-states/
   c. AUSTRALIA : https://www.legislation.gov.au/Details/C2016C00321
   d. GCC : https://www.saso.gov.sa/en/Laws-And-Regulations/Technical_regulations/Pages/default.aspx
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11. International Standards -
    a. Europe – https://www.en-standard.eu/
    b. USA - https://www.astm.org/Standards/F956.htm, https://www.astm.org/Standards/B36.htm
    c. Australia - https://www.standards.org.au/
    d. GULF - https://www.gso.org.sa/store
12. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

STATUES, FLOWER VASE, POOJA ITEMS, BRASS LAMPS & BRASS PRODUCTS, MORADABAD, MAHOBA

1. INTRODUCTION

Metal crafts are made by sculpting metals, mostly Zinc, Copper, Brass, Silver and Gold. The process of working with metals are used to create individual parts, assemblies, or large-scale structures. Metalworking is a science, art, hobby, industry and trade. Its historical roots span cultures, civilizations, and millennia. Metalworking has evolved from the discovery of smelting various ores, producing malleable and ductile metal, useful tools and adornments. Modern metalworking processes, though diverse and specialized, can be categorized as forming, cutting, or joining processes.

Moradabad is known as the 'Brass City' as it is renowned for brass work and has carved a niche for itself in the handicraft industry throughout the world. The designs made on the brass products here display culture, heritage, history and diversity. The patterns and designs used for decorating these items are inspired from a variety of sources, varying from Hindu Gods and Goddesses to paintings of the Mughal Era. The district has both small units as well as large industries engaged in the manufacture of metal goods. The work of washing, shaping and polishing handicraft metal items is carried out in the domestic units. The exporters in Moradabad have now begun to work with other metals like aluminium, stainless steel, iron etc.

Mahoba is also renowned for its prowess in the field of brassworks. The Artisans are known to have preserved designs for unique products such as lamp wick tongs which are not found in the modernized and bulk handling clusters.

The artefacts made of Copper, Brass, Zinc and Iron, are used in various parts of the country. Sculptures of religious importance, sculptures showing historical stories, decorative pieces are famous among these artefacts. Sculptures of Lord Shiva and Lord Ganesh made of Octal metal are popular metal works of Mahoba district. Now this industry has stepped up from cottage industry level and attained the level of small-scale industry. Modern technology has also entered in the industry along with traditional ways of manufacturing, reason being the quality and finish of products has been enhanced as world class. This industry provides employment to a significant people of the district.

The artefacts produced in Moradabad, have been awarded the GI tag in the year 2014, which has further cemented the position of the products in the global markets as a unique product.

2. RAW MATERIALS

The common raw materials for the creation of metal ware are:

- Brass, Bronze, Tin, Zinc, Lead and Iron
- Wax
- Sand
- Chemical binders
- Oil based clay

3. PRODUCTION PROCESS

The general process followed for manufacturing metal sculptures or other items is given below:

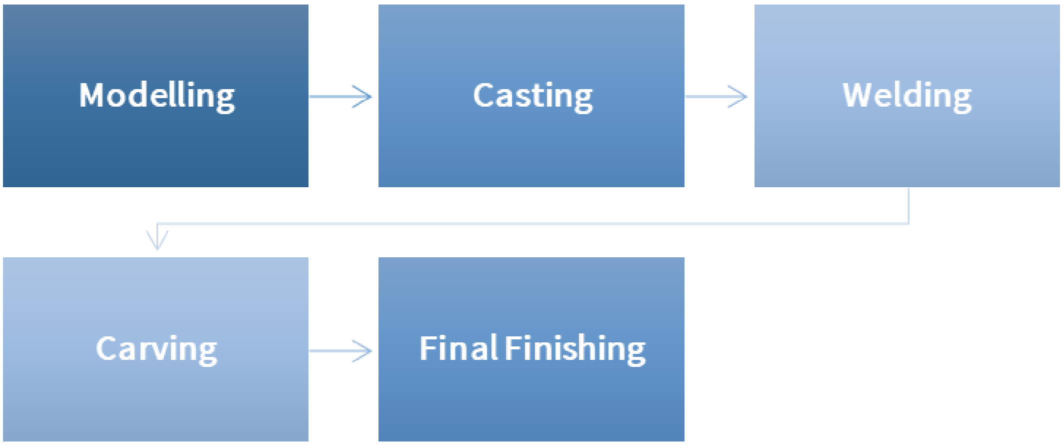

Figure 1: Flow chart for manufacturing

Mostly metal statues and sculptures are used for décor purposes. These are ornamental pieces that add on to the décor of the space. The mediums that are mostly used are brass, aluminium, bronze, copper, stainless steel and mild steel. Brass, however, is the preferred medium of expression of the artisans involved in this trade and are known to produce exquisite articles using traditional craft methodology. The most common method of manufacturing the products can be described as below:

3.1. Hammering and Jointing of Plates

a. **Cutting of the Brass Sheet:** The sheet of required thickness is procured from the market. It is first straightened and then with the help of a scale, marker and compass, markings are done on the metal sheet for the product to be made. In the next stage, the sheet is cut either by chisel (if thick) or by metal scissor (if thin). The shape can be square or circular depending on the final product to be made. The thickness of the metal also depends on the final product.

b. **Shaping of the product:** Once the sheet is cut, it is placed on a *Bangad* (a solid metal ring of iron on which the metal sheet is placed and beaten with the hammer along with simultaneously rotating it) and hammered by wooden mallets into shape. Traditionally, *Thiya* (a solid metal stand) is also used to shape the product or to make the neck of the vessels. While shaping the metal, the sheet is rotated continuously for an even treatment of the shape of metal body. For vessels such as pots, welding of the metal pieces made by hammering the sheet is needed.

c. **Joining of pieces:** The products are made in parts and finally joined either by heating or by gas welding. Sometimes, artisans also join two pieces with traditional interlocking of the metals. These joints are converted into a design element by further heating and beating them. This process is carried out by heating the metal pieces in Bhatti at high temperatures. This makes the metal more flexible, facilitating blending into each other. In today's times, the artisans use gas welding as it is less time consuming and easy. For this process, both the pieces are heated till they become red hot. On the surface, *Suhaga* (a powder used in welding) is applied and a thin brass rod is melted which joins the two pieces. The gaps are filled by melting the brass rod on the product, thus helping in joining the two pieces.

d. **Finishing:** Once the product is joined, the excess metal accumulated on the surface due to welding is removed by a grinding machine or files. These items then go through acid wash to clean the surface. The product is then buffed using buffs of different grades/numbers.

3.2. Sand Box Casting Method

a. **Making the mould:** The first step is to make the mould or 'master-piece' from which multiple products would be replicated (cast). These are usually made of wax, as it is soft and easy to work with. Sometimes wood is also used. This 'master-copy' is always created in two (or more) detachable halves to make sand casting convenient.

b. **Melting the metal:** The brass metal ingots or alloys (Copper, Zinc etc.) are heated in a furnace to melt them.

c. **Casting:** Sand casting is the traditional method of making brass ware. Sand is used in the two halves of the mould box to cast the metal which is prepared by packing sand (locally called '*masala'*) around the 'master-copy' tightly. The chemical binder in the sand aids in holding the shape of the mould after which it is removed and molten metal is poured in its cavity. After being allowed to cool for few minutes, the cast metal is removed from the mould box. The sand and the 'gating' (pathway made in the mould for directing the flow of molten metal into the cavity) are broken from the cast using a hammer. These are re-used in the next casting.

d. **Scraping:** The products are mounted on a lathe and scraped to remove any protruding materials and to smoothen the surface. If the products are made in two parts (pots etc.) then the two pieces are joined by welding followed by scraping and polishing.

e. **Engraving:** Engraving is the most refined and artistic of all the processes. The design that has to be engraved, is first sketched on paper and then scaled-up according to the size of the product. Measurement is an important aspect to make the pattern look harmonious and uniform. These are mainly inspired from different forms of nature like trees, flowers, birds and animals. The geometric ones have their influence from the Mughal architecture. First, an outline of the whole design is done with a fine engraving tool hammered with a wooden block and then broader tools are used to engrave the background and give depth to the pattern. These are often filled with colourful lac or enamel.

f. **Polishing:** Polishing mainly includes cleaning the brass ware with a soft scrub and then buffing it on the machine for golden sheen.

3.3. Lost Wax Casting Method

Lost wax casting is also known as precision casting and is widely used for metal sculptures. It is a cire-perdue method where the molten metal is poured into a mould which is developed using a wax model to get the desired shape. Different types of finishes can be given to create different looks to the finished artefact. The major process steps are:

a. **Model Making:** An exact replica of the existing sculpture or artwork is made using primarily wax. Other materials such as an oil based clay may be used too.

b. **Mould Making:** A mould is made usually in two halves of the original wax model. The inner mould is the exact negative of the original model. Wax is then poured into this mould and swished around to coat the sides of the mould evenly.

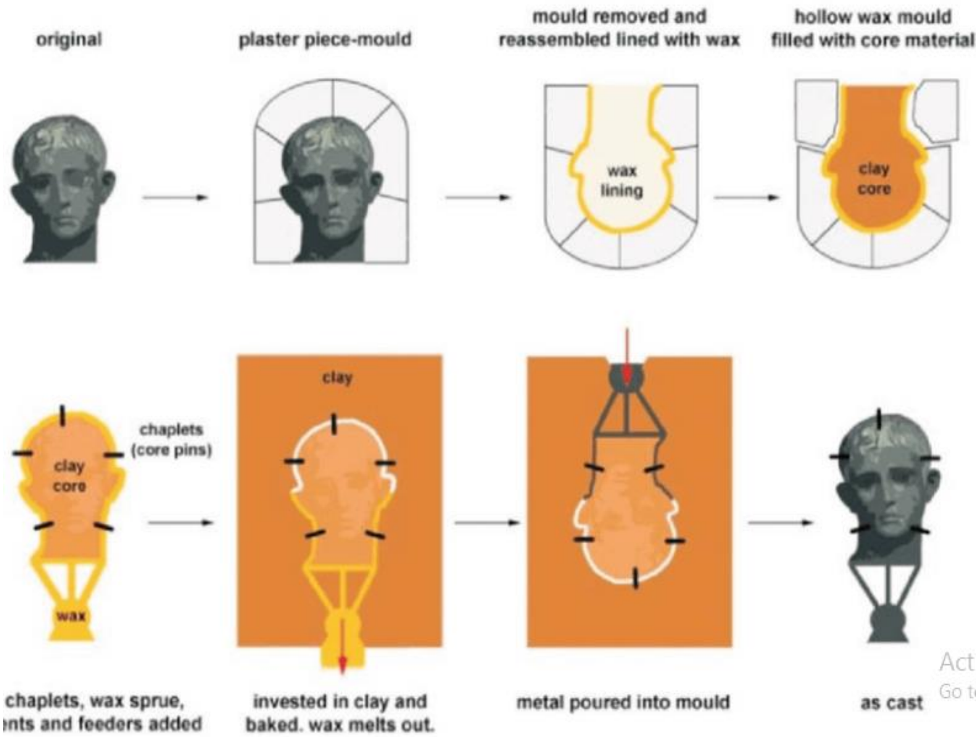

Figure 2: Lost wax casting method

c. **Spruing:** After the hollow wax halves are removed from the mould, finishing is accomplished by a process called 'chasing' to remove the marks left by the mould. The two halves are heated and joined together. Foundries often use registration marks to indicate exactly where they would go. This wax copy is *sprued* with a tree like structure of wax that eventually provides paths for the molten casting metal to flow and for air to escape.

d. **Slurry Coating:** The readied piece is dipped into a slurry of silica and then in a sand like stucco. This slurry and sand mixture is called a ceramic shell mould material (though it is not actually ceramic) in the industry parlance. This shell is allowed to dry and the process is repeated till the required thickness of the external coating has been achieved.

e. **Burnout:** The coated shell is heated in a kiln, which causes the wax inside to melt. This usually gets burnt up, lending its name to the process *"Lost wax method*". Only the negative space remains within the mould.

f. **Pouring:** The mould is heated again in the kiln and molten metal is poured in. It is important that there should be no temperature difference between the moulds and metal otherwise the mould would shatter.

g. **Finishing:** The metal is allowed to cool, and then the external mould is broken. The sprues which are attached to the cast shape are cut off and the entire object is finished by removal of all rough edges etc. and polished.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check the fabricated vessel/artefact for workmanship
- Check for the uniformity of shape and dimensions
- Check for finishing of protective coatings and other damages, if any
- Check the welding joints

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulation (EC)1907/2006	a. European Commission b. European Chemical Agency (ECHA)	a. Composition of alloy b. Reaction when in contact with food material c. Compliance with the specific release limits(SRLs)

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
2	United States of America	**a.** Department of Health and Human Services (HHS) regulations. **b.** Consumer Product Safety Improvement Act (CPSIA), Act of 2008 **c.** Consumer Product Safety Act (CPSA), Act of 1972 **d.** Federal Hazardous Substances Act (FHSA), Act of 1960. **e.** California Proposition 65, Act of 1986	**a.** Food and Drug Administration (FDA) **b.** Consumer Products Safety Commission **c.** California State	**a.** Migration of harmful substances **b.** Durability and corrosion resistance **c.** Scratch resistance **d.** Specifies requirements with respect to Chemicals and heavy metals **e.** Specifies requirement for suitable product labelling **f.** Specifies banned substances **g.** Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products
3	Australia	**a.** Commerce (Trade Description) Act, 1905	**a.** Australian Customs, Office of the Comptroller General	**a.** Specifies the requirements for Mandatory Labelling of products
4	Middle East	**a.** Saudi Standards, Metrology and Quality Organization (SASO) regulations	**a.** Safety, Saudi Standards, Metrology and Quality Organization (SASO)	**a.** General product safety regulation **b.** Labelling Requirement

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Scope**	**Region/ Country**	**Standard**	**Scope**
General (Refer Table 1)	**IS 3288: Part 1: 1986 (Reaffirmed Year: 2021)**	Glossary of terms used for material in the field of copper and copper alloys	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 5047: Part 1: 1986 (Reaffirmed Year: 2022)**	Covers the requirements of cast aluminium and its alloys in the form of ingots and castings for general engineering purposes	European Union	**BS EN 1981 - 2003**	Master alloys: Copper and copper alloys
	IS 3685: 1966 (Reaffirmed Year: 2018)	Methods to determine copper and other metals in brass	USA	**ASTM E 716-16**	Sampling and sample preparation of aluminium and aluminium alloys for determination of chemical composition

Domestic Standards			International Standards		
Stage	**Code**	**Scope**	**Region/ Country**	**Standard**	**Scope**
	IS 737: 2008 (Reaffirmed Year: 2018)	Alloy and wrought aluminium sheets	USA	**ASTM B0036M**	Standard specification for brass plate, sheet, strip, and rolled bar
	IS 737: 1981 (Reaffirmed Year: 2021)	High tensile brass ingots and castings	USA	**ASTM B209M-14**	Standard specification for aluminium and aluminium-Alloy sheet and plate (metric)
	IS 2676: 1981 (Reaffirmed Year: 2022)	Wrought aluminium, aluminium alloys sheet and strips dimensions	Australia	**AS 4861 - 2004**	Aluminium and aluminium alloys - Determination of impurities and alloying elements - Atomic emission spectrometric method
	-	-	Australia	**AS 1566 - 1997**	Copper and copper alloys - Rolled flat products
	-	-	Middle East	**GSO ISO 209:2008**	Chemical composition of aluminium and alloys
	-	-	Middle East	**GSO ASTM B148:2010**	Aluminium-Bronze sand castings
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 1387: 1993 (Reaffirmed Year: 2018)**	Requirements of three grades of brass rods, namely, CuZn20, CuZnSO and CuZn40 for general engineering purposes	European Union	**BS EN 12167: 2016**	Copper and copper alloys. Profiles and bars for general purposes.
	IS 1599: 2019	Metallic materials - Bend test (Fourth revision)	European Union	**BS EN 1982: 2017**	Copper and copper alloys. Ingots and castings
	IS 3052 : 1986 (Reaffirmed Year: 2021)	Dimensions and tolerances for wrought copper and copper alloy sheet, strip and foil for general engineering purposes	Middle East	**GSO ISO 10308: 2013**	Metallic coatings - Review of porosity tests
	IS 407: 1981 (Reaffirmed Year: 2021)	Brass tubes for general purposes	-	-	-
	IS 304: 1981 (Reaffirmed Year: 2021)	High tensile brass ingots and castings	-	-	-
Finished Goods (Refer Table 4.a & 4.b)	**IS 5523: 1983 (Reaffirmed Year: 2021)**	Testing anodic coatings on aluminium and its alloys	European Union	**BS EN ISO 6509-1: 2014**	Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc test method
	-	-	USA	**ASTM B605-95a**	Standard specification for electrodeposited coatings of tin-nickel alloy

Domestic Standards			International Standards		
Stage	Code	Scope	Region/ Country	Standard	Scope
	-	-	USA	**ASTM A751 - 20**	Standard test methods and practices for chemical analysis of steel products

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 3288: Part 1: 1986 (Reaffirmed Year: 2021)** - Glossary of terms relating to copper and copper alloys	▪ Glossary of terms used for material in the field of copper and copper alloys

6.2. Table 2.a: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 737: 2008 (Reaffirmed Year: 2018)** - Alloy and wrought aluminium sheets	▪ Chemical composition of wrought and aluminium alloy sheets ▪ Mechanical properties ▪ Comparison of ISO and IS designations ▪ Characteristics and typical uses
	IS 2676: 1981 (Reaffirmed Year: 2022) - Wrought aluminium, aluminium alloys sheet and strips dimensions	▪ Dimensions for wrought aluminium and aluminium alloys. ▪ Specification of tolerances ▪ Thickness of wrought aluminium and aluminium alloy sheets ▪ Shearing tolerances
	IS 5047: Part 1: 1986 (Reaffirmed Year: 2022) - Light metals and their alloys	▪ Commonly used terms relating to unwrought aluminium and both main and special types of wrought aluminium
	IS 304: 1981 (Reaffirmed Year: 2021) - High tensile brass ingots and castings	▪ Standard covers the requirements for two grades of high tensile brass ingots for re-melting purposes and castings
	IS 3685: 1966 (Reaffirmed Year: 2018) - Methods of chemical analysis of brasses	▪ Methods for determination of copper, lead, tin, manganese, phosphorus, nickel, iron, silicon, aluminium, zinc, and arsenic in the ranges as specified in the relevant Indian Standards on brasses ▪ Quality of reagents ▪ Methodology ▪ Referenced documents

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**BS EN 1981 - 2003** - Copper and copper alloys	▪ Master alloys: copper and copper alloys
	ASTM B0036M - Standard specification for brass plate, sheet, strip, and rolled bar	▪ Requirements brass plates, sheets, strips, and rolled bars ▪ The standard tempers of as hot rolled, rolled, annealed, and annealed-to-temper materials are detailed in this specification ▪ Units of measurements

Stage	Code	Scope
	ASTM E 716-16 - Standard practices for sampling and sample preparation of aluminium and aluminium alloys for determination of chemical composition by spark atomic emission spectrometry	▪ Procedures for casting a chill cast disc from molten aluminium during the production process ▪ Procedures for producing a chill cast disk sample from molten metal produced by melting pieces cut from products
	ASTM B209M-14 - Standard specification for aluminium and aluminium-alloy sheet and plate (metric)	▪ Specifications metric for aluminium and aluminium-alloy sheet and plate (metric)
	AS 4861 – 2004 - Determination of impurities and alloying elements	▪ Determination of impurities and alloying elements
	AS 1566 – 1997 - Copper and copper alloys - Rolled flat products	▪ Copper and copper alloys - Rolled flat products

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 1387: 1993 (Reaffirmed Year: 2018)** - Requirements of three grades of brass rods	▪ Standard covers the general requirements for the supply of metallurgical materials including materials for metallurgical industries and finished metal products
	IS 1599: 2019 - Metallic materials - Bend test (Fourth revision)	▪ Metallic materials - Bend Test
	IS 3052 : 1986 (Reaffirmed Year: 2021) - Copper and copper alloys	▪ Dimensions and tolerances for wrought copper and copper alloy sheet, strip and foil for general engineering purposes
	IS 304: 1981 (Reaffirmed Year: 2021) - High tensile brass ingots and castings	▪ Dimensions for wrought aluminium and aluminium alloys ▪ Specification of tolerances ▪ Thickness of wrought aluminium and aluminium alloy sheets ▪ Shearing tolerances
	IS 407: 1981 (Reaffirmed Year: 2021) - Brass tubes for general purposes	▪ Solid drawn brass tubes requirements for general purposes: Chemical composition ▪ Referenced standards ▪ Terminology and supply of material conditions

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS EN 12167: 2016** - Copper and copper alloys. Profiles and bars for general purposes	▪ Specifies the composition, property requirements and dimensional tolerances for copper alloy profiles including L-, T-, U-shaped cross-sections, and bars, finally produced by drawing or extruding
	BS EN 1982: 2017 - Copper and copper alloys. Ingots and castings	▪ Composition, mechanical properties and other relevant characteristics of copper and copper alloys ▪ Sampling methods ▪ Test Methods

Stage	Code	Scope
	GSO ISO 10308: 2013 - Metallic coatings - Review of porosity tests	▪ Methods for revealing pores (see ISO 2080) and discontinuities in coatings of aluminium, anodized aluminium, brass, cadmium, chromium, cobalt, copper, gold, indium, lead, nickel, nickel-boron, nickel-cobalt, nickel-iron, nickel-phosphorus, palladium, platinum, vitreous or porcelain enamel, rhodium, silver, tin, tin-lead, tin-nickel, tin-zinc, zinc and chromate or phosphate conversion coatings (including associated organic films) on aluminium, beryllium-copper, brass, copper, iron, NiFeCo alloys, magnesium, nickel, nickel-boron, nickel-phosphorus, phosphor-bronze, silver, steel, tin-nickel and zinc alloy basis metal

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 5523: 1983 (Reaffirmed Year: 2021)** - Methods of testing anodic coatings on aluminium and its alloys	▪ Determination of thickness of anodized coating on aluminium and its alloys and methods of testing ▪ Methods of determination of resistance to abrasion ▪ Methods of testing for light fastness and allocation of light fastness number ▪ Methods for testing of sealing

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN ISO 6509-1: 2014** - Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc test method	▪ Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc test method
	ASTM B605-95a - *Standard specification for electrodeposited coatings of tin-nickel alloy*	▪ Standard specification for electrodeposited coatings of tin-nickel alloy
	ASTM A751 – 20 - Standard Test Methods and Practices for Chemical Analysis of Steel Products	▪ Test methodologies wet and instrumental ▪ Referenced documents

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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   b. USA: https://www.compliancegate.com/
   c. AUS: https://www.compliancegate.com/
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16. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

1. INTRODUCTION

Brass is non-magnetic and a good conductor of heat. It is both durable and easy to work on. All these properties make it ideal for being made into and used as utensils in kitchens.

The brassware of Moradabad is an ancient craft and the city is known as the "City of Brass". The artisans engaged in this craft are known to make various types of artistic utensils and pieces such as spoons, decorative brassware and more. Brass handicrafts produced here are popular world over and they represent the true Indian craftsmanship and the old grandeur for which Indian Brass ware is well known. These exclusive and elegantly crafted brassware are not only popular in national markets but are also in huge demand in international markets.

Brass is an alloy of copper (60%-70%) and zinc (40%-30%); while copper is reddish in colour, brass is yellowish because of presence of zinc. Brass is more malleable than bronze or zinc which allows it to be easily drawn into sheets. It is non-magnetic, good conductor of heat, durable and easy to work on. All these properties make it ideal for being made into and used as utensils in kitchens.

India and the world, is fast realising that traditional Indian cooking methods were perhaps the best for an individual's health. The types of utensils that were used in the earlier times, which were essentially metal or clay based, also contributed a lot to maintaining good health.

The brassware produced in Moradabad has been bestowed by the GI Tag, which has helped to further enhance its reputation in the International markets.

2. RAW MATERIALS

The raw materials commonly used for making brass utensils are as below:

- Brass (alloys of copper, silver) ingots or sheets
- Brazing flux powder

3. MANUFACTURING PROCESS

3.1. Sand Box Casting Method

a. **Making the mould:** The first step is to make the mould or 'master-piece' from which multiple products would be replicated (cast). These are usually made of wax, as it is soft and easy to work with. Sometimes, wood is also used. This 'master-copy' is always created in two (or more) detachable halves to make sand casting convenient.

b. **Melting the metal:** The brass metal ingots or alloys (Copper, Zinc etc.) are heated in a furnace to melt them.

c. **Casting:** Sand casting is the traditional method of making brass ware. Sand is used in the two halves of the mould box to cast the metal which is prepared by packing sand (locally called 'masala') around the 'master-copy' tightly. The chemical binder in the sand aids in holding the shape of the mould after which it is removed and molten metal poured in its cavity. After being allowed to cool, the cast metal is removed from the mould box. The sand and the 'gating' (path way made in the mould for directing the flow of molten metal into the cavity) are broken from the cast using a hammer. These are re-used in the next casting.

d. **Scraping:** The products are mounted on a lathe and scraped to remove any protruding materials and to smoothen the surface. If the products are made in two parts (pots etc.) then the two pieces are joined by welding and then the product is subjected to scraping and polishing.

e. **Engraving:** Engraving is the most refined and artistic of all the processes. The design that has to be engraved is first sketched on paper and then scaled-up according to the size of the product. Measurement is an important aspect to make the pattern look harmonious. First, an outline of the whole design is made with a fine engraving tool hammered with a wooden block and then, broader tools are used to engrave the background and give depth to the pattern. These are often filled with colourful lac or enamel.

f. **Polishing:** Polishing mainly includes cleaning the brassware with a soft scrub and then buffing it on the machine for golden sheen.

g. **Qalai or Kalai:** For utensils to be used for cooking, a layer of tin is deposited to coat the surface. This is known as 'Qalai' locally. This is an important process as the unprotected copper reacts with moisture present in the air and forms copper carbonate (which can be seen on untreated copper vessels as light green rust on surface) which is poisonous and can be harmful if mixed with food. This layer of tin protects the copper vessel below. This however, wears off with time and as a result the copper vessels have to be repeatedly treated with the process of depositing a layer of tin. This is one of the reasons that has led to the decline in the use of copper vessels with the advent of aluminium and stainless-steel vessels.

Figure 1: Making of Mould

Figure 2: Pouring of brass

Figure 3: Engraving

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for uniformity in product or any other visual defects such as cracks, chipping, deformities etc.
- Check for smoothness & finishing of welding joints.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce. (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	**a.** Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011	**a.** FSSAI	**a.** Governs use of Food grade utensils apart from other food safety standards **b.** All equipment and utensils which come in contact with food.

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	**a.** Commission Regulation (EC) No. 2023/2006 of 22 December 2006 on good manufacturing practice for materials and articles intended to come into contact with food. **b.** Framework Regulation (EC) No. 1935/2004.	**a.** European Commission	**a.** Specifies requirements with respect to Chemicals and heavy metals **b.** Specifies Good manufacturing practices for materials coming in contact with food **c.** Specifies General safety requirements for all food contact materials. In addition, specific measures have been introduced to plastics
2	United States of America	**a.** Consumer Product Safety Improvement Act (CPSIA), Act of 2008. **b.** Consumer Product Safety Act (CPSA), Act of 1972. **c.** Federal Hazardous Substances Act (FHSA), Act of 1960. **d.** California Proposition 65, Act of 1986. **e.** Code of Federal Regulations, Section 21	**a.** California State **b.** Consumer Products Safety Commission **c.** Food and Drug Administration (FDA)	**a.** Specifies requirements with respect to Chemicals and heavy metals **b.** Specifies requirement for suitable Product Labelling **c.** Specifies Banned substances **d.** Any manufacturing unit having more than ten employees are required to label products if they contain carcinogenic products

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
				e. Regulations of hazardous substances in food contact materials (FCM)
3	Middle East	a. GSO Technical Regulations ▪ GSO 2231: 2012 ▪ GSO 839/1997	a. GCC Standardization Organization (GSO)	a. General requirements for the materials intended to come into contact with food

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 3288: Part 1: 1986**	Glossary of terms relating to copper and copper alloys: Part 1 Materials	European Union	**BS EN 1655: 1997**	Copper and copper alloys. Declarations of conformity
	IS 1387: 1993 (Reaffirmed Year: 2018)	General requirements for the supply of metallurgical materials	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	**IS 422: 1981 (Reaffirmed Year: 2021)**	Brass sheet and strip for the manufacture of utensils	European Union	**BS EN 1981 - 2003**	Copper and copper alloys. Master alloys
	-	-	USA	**ASTM B0036M**	Standard specification for brass plate, sheet, strip, and rolled bar
	-	-	USA	**ASTM B0455M**	Standard Specification for Copper-Zinc-Lead Alloy (Leaded-Brass) Extruded Shapes
	-	-	Australia	**AS 1566 - 1997**	Rolled flat products
Manufacturing/ Processing (Refer Table 3.a & 3.b)	**IS 4822: 1981 (Reaffirmed Year: 2021)**	Specifications for brass cooking utensils	European Union	**BS 3788**	Specification for tin coated finish on culinary utensils
	IS 3655: 1985 (Reaffirmed Year: 2021)	Recommended practices for electroplating	European Union	**BS EN 12167: 2016**	Copper and copper alloys. Profiles and bars for general purposes
	IS 16286: 2014 (Reaffirmed Year: 2020)	Spoon specifications with brass as parent material	European Union	**BS EN 1982: 2017**	Copper and copper alloys. Ingots and castings
	-	-	Middle East	**GSO ISO 10308: 2013**	Metallic coating
Finishing (Refer Table 4.a & 4.b)	**IS 3685: 1966 (Reaffirmed Year: 2018)**	Methods of chemical analysis of brasses	European Union	**BS EN ISO 6509-1: 2014**	Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc - Test method
	IS 1501: Part 1: 2020	Metallic materials — Vickers hardness Test Part 1 Test Method (Fifth Revision)	USA	**ASTM B605-95a**	Standard specification for electrodeposited coatings of tin-nickel alloy

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	-	-	Australia	**AS 1515.4-1978 Rec - 2016**	Methods for analysis of copper alloys
	-	-	Australia	**AS 2345 - 2006**	Dezincification resistance of copper alloys
	-	-	Middle East	**GSO ISO 6509: 2013**	Corrosion of metals and alloys

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 3288: Part 1: 1986** - Glossary of terms relating to copper and copper alloys	▪ Glossary of terms and technical definitions
	IS 1387: 1993 (Reaffirmed Year: 2018) - General requirements for the supply of metallurgical materials	▪ Raw materials supply for metallurgical industries ▪ Terminology and Inspection, sampling
	BS EN 1655: 1997 - Copper and copper alloys-Declarations of conformity	▪ Declaration of conformity, copper and copper alloys

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 422: 1981 (Reaffirmed Year: 2021)** - Specification of brass sheet and strip for the manufacture of utensils	▪ Chemical composition ▪ Adjunct standards ▪ Physical properties

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**BS EN 1981 – 2003** - Copper and copper alloys	▪ Master Alloys: Copper and copper alloys
	ASTM B0036M - Standard specification for brass plate, sheet, strip, and rolled bar	▪ Requirements brass plates, sheets, strips, and rolled bars ▪ The standard tempers of as hot rolled, rolled, annealed, and annealed-to-temper materials are detailed in this specification ▪ Units of measurements

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 4822: 1981 (Reaffirmed Year: 2021)** - Specification for brass cooking instruments	▪ Requirements of brass cooking instruments ▪ Materials and dimensions
	IS 3655: 1985 (Reaffirmed Year: 2021) - Recommended standard practice for electroplating	▪ Recommended methods of practices for electroplating ▪ Process for parts in bulk ▪ Plating baths ▪ Adjunct standards
	IS 16286: 2014 (Reaffirmed Year: 2020) - Spoons specifications	▪ Requirements for the following types of spoons made from stainless steel and nickel silver plated with brass as a parent material

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	**BS 3788** - Specification for tin coated finish on culinary utensils	▪ Specification for tin coated finish on culinary utensils
	ASTM B0455M - Standard specification for copper-zinc-lead alloy (leaded-brass) extruded shape	▪ Requirements for extruded leaded-brass angles, channels, and other architectural shapes of solid cross section produced in copper alloy ▪ Exclusions ▪ Units of measurements
	AS 1566 – 1997 - Copper and copper alloys - Rolled flat products	▪ Copper and copper alloys - Rolled flat products
	BS EN 12167: 2016 - Copper and copper alloys. Profiles and bars for general purposes	▪ Specifies the composition, property requirements and dimensional tolerances for copper alloy profiles including L-, T-, U-shaped cross-sections, and bars, finally produced by drawing or extruding
	BS EN 1982: 2017 - Copper and copper alloys. Ingots and castings	▪ Composition, mechanical properties and other relevant characteristics of copper and copper alloys ▪ Sampling methods ▪ Test Methods
	GSO ISO 10308: 2013 - Metallic coatings - Review of porosity tests	▪ Methods for revealing pores (see ISO 2080) and discontinuities in coatings of aluminium, anodized aluminium, brass, cadmium, chromium, cobalt, copper, gold, indium, lead, nickel, nickel-boron, nickel-cobalt, nickel-iron, nickel-phosphorus, palladium, platinum, vitreous or porcelain enamel, rhodium, silver, tin, tin-lead, tin-nickel, tin-zinc, zinc and chromate or phosphate conversion coatings (including associated organic films) on aluminium, beryllium-copper, brass, copper, iron, nifeco alloys, magnesium, nickel, nickel-boron, nickel-phosphorus, phosphor-bronze, silver, steel, tin-nickel and zinc alloy basis metal

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 3685: 1966 (Reaffirmed Year: 2018)** - Methods of chemical analysis of brasses	▪ Sampling ▪ Quality of reagents, Testing methodologies ▪ Test methodology
	IS 1501: Part 1: 2020 - Vickers hardness test Part 1 Test method	▪ Hardness tests

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**BS EN ISO 6509-1: 2014** - Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc Test method	▪ Corrosion of metals and alloys. Determination of dezincification resistance of copper alloys with zinc Test method
	ASTM B605-95a - *Standard specification for electrodeposited coatings of tin-nickel alloy*	▪ Standard specification for electrodeposited coatings of tin-nickel alloy
	AS 1515.4-1978 Rec – 2016 - Methods for the analysis of copper alloys	▪ Methods for the analysis of copper alloys. Electrolytic determination of copper in wrought and cast copper alloys ▪ Part 4: Electrolytic determination of copper in wrought and cast copper alloys
	AS 2345 – 2006 - Dezincification resistance of copper alloys	▪ Dezincification resistance of copper alloys
	GSO ISO 6509: 2013 - Corrosion of metals and alloys	▪ Determination of dezincification resistance of brass. Applicable to brass exposed to fresh or saline water.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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1. Product information: http://odopup.in/en/article/moradabad
2. Product information: https://en.wikipedia.org/wiki/Brass
3. Product Information: https://www.finishing.com/558/17.shtml
4. Domestic Legislation/Regulation:
   https://fssai.gov.in/upload/uploadfiles/files/Licensing_Regulations.pdf
   https://foodsafetyhelpline.com/licensing-registration-of-food-businesses/
5. Mandatory IS mark: https://bis.gov.in/index.php/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/
6. Indian Standards under Compulsory Certification: https://www.bsbedge.com/mandatory-indian-standards
7. International Legislation/Regulation:
   a. EU: https://www.legislation.gov.uk/eur/2006/2023/contents
      - https://www.legislation.gov.uk/uksi/2012/2619/contents/made
   b. USA: https://www.compliancegate.com/heavy-metal-regulations-united-states/
      - https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=111&showFR=1&subpartNode=21:2.0.1.1.11.4
   c. AUS: https://www.compliancegate.com/food-contact-material-regulations-australia
   d. GULF (GCC): https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation/
8. Domestic Standards for Brass Utensils and raw materials:
   https://standardsbis.bsbedge.com/
9. BIS product conformation overview: https://bis.gov.in/index.php/product-certification/product-certification-overview/
10. BIS Certified Labs for Testing Products as per IS Standards:
    http://164.100.105.198:8096/bis_access/iswise_v2.html
11. Mandatory BIS standards: https://www.bsbedge.com/mandatory-indian-standards
12. ISO Standards: https://www.iso.org/obp/ui#search
13. International Standards:
    a. EU: https://www.en-standard.eu/
    b. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
    c. AUS: https://www.standards.org.au/
    d. GULF: https://www.gso.org.sa/store/
14. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
http://nabcb.qci.org.in/

APPENDIX II: IMPORTANT INSTITUTIONS

METALCRAFT			
Institute	**Areas of Expertise**	**Postal Address**	**Contact**
Process and Product Development Centre (PPDC)	▪ Material Testing ▪ Consultancy ▪ Product Development ▪ Training	Process and Product Development Centre (PPDC), Foundry Nagar, Agra, Uttar Pradesh - 282006	Email: tcppdcagra@dcmsme.gov.in Phone: +91–562- 2344673 Website: http://www.ppdcagra.dcmsme.gov.in/

ONE DISTRICT ONE PRODUCT

UTTAR PRADESH

ODOP Cell, Directorate of Industries, Niryat Bhawan, First Floor, 8 Cantt Road, Qaiserbagh, Lucknow – 226001